COURSE DESCRIPTION FORM: MT-1004 Linear Algebra

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

PROGRAMs TO BE EVALUATED BS-CS – **Fall 2023**

Course Description

(Fill out the following table for each course in your computer science curriculum. A filled out form should not be more than 2-3 pages.)

Course Code	MT-1004
Course Title	Linear Algebra
Credit Hours	3
Prerequisites by Course(s) and Topics	Nil
Grading Policy	Absolute Grading
Policy about missed assessment items in the course	Retake of missed assessment items (other than midterm/ final exam) will not be held. For a missed midterm/ final exam, an exam retake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decide the exam retake cases.
Course Plagiarism Policy	Plagiarism in project or midterm/ final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	**100% Theory**

Assessment Item	**Number**	**Weight (%)**
Assignments and HomeWorks	16	08
Quiz	10	12
Sessional I & II	1	15 (Each)
Final Exam	1	50

Course Instructors	Muhammed Ali, Sara Aziz, Irfanullah, Zaira.
Lab Instructors (if any)	
Course Coordinator	Muhammed Ali
URL (if any)	
Current Catalog Description	Basic ideas and techniques to understand linear systems and their interpretation in terms of solutions. Vector-spaces and its properties, Eigen decomposition, linear transformations, orthogonality (Gram-Schmidt process) and least square solution to a system.

Textbook	Linear Algebra and its Applications (4th Edition) by David C. Lay.
Reference Material	**Elementary Linear Algebra,** by Howard Anton and Chris Rorres, John Wiley, 11th Edition.
Course Learning Outcomes	**A. Course Learning Outcomes (CLOs)** At the completion of the course, the students shall be able to: 1. Solve the system of linear equations. 2. Analyse properties of vector spaces. 3. Perform matrix transformations. 4. Apply Eigenvalue decomposition to a matrix.

B. Program Learning Outcomes	
For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.	
1. Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.	
2. Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.	
3. Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	
4. Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations	
5. Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.	
6. Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports,	

design documentation, make effective presentations, and give and understand clear instructions.	
7. Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.	
8. Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.	
9. Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.	

C. Mapping of CLOs on PLOs
(CLO: Course Learning Outcome, PLOs: Program Learning Outcome

		PLOs											
		1	2	3	4	5	6	7	8	9	10	11	12
CLOs	1	✔											
	2	✔											
	3	✔											
	4	✔											

Topics covered in the course with number of lectures on each topic (assume 15 weeks of instruction and 1.5 hours lecture duration)

Topics	Lectures
System of Linear Equations & its solution. Applications of system of linear equations. Linear Transformation and its matrices. Introduction to multilinear maps, bilinear maps.	9
Elementary Matrices. Inverse of a matrix. Brief introduction of determinants and its applications.	6

	Field & Vector Space, Sub-spaces. Basis and Dimension of vector space, Null space, Row space, Column spaces and its properties.	9
	Eigen-Decomposition. Solution for System of ODE's with the help of Eigen values & Eigenvectors.	6
	Affine combination, affine independence, affine transformation. Inner product. Orthogonal and orthonormal vectors, Gram-Schmidt process for finding orthonormal basis. Pseudo inverse and Penrose Inverse of general matrices.	8
	Orthogonality. Gram-Schmidt process & QR-decomposition, and least square solution of linear system, Positive Definite Matrices, Singular Value Decomposition	7

Laboratory Projects/Experiments Done in the Course				
Programming Assignments Done in the Course				
Class Time Spent per Week (in percentage)	**Theory (%)**	**Problem Analysis (%)**	**Solution Design (%)**	**Social and Ethical Issues (%)**
	10	20	60	10
Oral and Written Communications				

COURSE CONTENTS

Weeks	Contents/ Topics	Courseware Events (Lab/ Case Study/ Quiz/ Assignment/.)	Comments
Week-01	**Introduction to system of linear equations. (Ex 1.1)** **Echelon form & Reduced Echelon form of a matrix. (Ex 1.2)**	H.W # 1	
Week-02	**Consistent & In-consistent System. System of homogenous Equation and its solution. (EX 1.3-1.6)**	H.W # 2 Quiz #1	
Week-03	**Applications of system of linear equations (Network flow problems, Balancing chemical equations, Polynomial Interpolations etc) (Ex 1.7). Linear transformations. Matrix of linear transformation. Composition of linear transformations. (Ex 1.8-1.9)**	H.W #3 Quiz # 2	

Week-04	**Elementary Matrices. Decomposition with elementary matrices. Inverse of a matrix with help of elementary matrices. Invertibility and solution of homogenous / non-homogenous equations. Diagonal and Symmetric Matrices and their properties. (Ex 2.1-2.4)**	H.W # 4 Quiz # 3 Assignment # 01	
Week-05	**Determinants and its applications. (Ex 3.1-3.4). Introduction to vector spaces and related examples. Sub-spaces and related examples. (Ex 4.1-4.2)**	H.W #5	
Week-06	**Introduction to multilinear maps, bilinear maps as tensors of rank 2, symmetric and alternating tensors, tensor product and wedge product (Book of Loring W. Tu. Section 3 chapter 1).**	H.W # 6 Quiz # 4	
Week-07	**Linear independence and spanning set. Basis of vector space. Change of Basis. (Ex4.3-4.7)**	H.W # 7 Quiz #5	
Week-08	**Null space, Row space, Column spaces and their properties.**	H.W # 8 Quiz # 6	
Week-09	**Eigenvalues & Eigenvectors, Diagonalization of a square matrix and its uses. (Ex 5.1-5.3)**	H.W # 9	
Week-10	**Discrete Dynamical Systems & Solution for System of ODE's with the help of Eigenvalues & Eigenvectors. (Ex 5.5-5.6)**	H.W # 10	
Week-11	**General linear transformations. Kernel and Range of linear transformation. One-to-one and onto Linear Transformations. Eigenvectors and linear transformation. (Ex 5.4)**	H.W #11 Quiz # 7	
Week-12	**Affine combination, affine independence, affine transformation (Section 8.1, 8.2 and 8.3)**	H.W # 12 Quiz # 8	
Week-13	**Inner product. Orthogonal and orthonormal vectors, Gram-Schmidt process for finding orthonormal basis. (Ex 6.1-6.3)**	H.W # 13 Quiz # 9 Assignment # 02	
Week-14	**QR-Decomposition (Ex 6.4) Least square method to find best approximate solution of an inconsistent system. (Ex 6.5). Pseudo inverse and Penrose Inverse of general matrices.**	H.W # 14 Quiz # 10	
Week-15	**Orthogonal Decomposition & Quadratic Form (Ex 7.1-7.2) Constraint Optimization. (Ex 7.3)**	H.W # 15 Quiz # 11	
Week-16	**Positive Definite Matrices, Singular Value Decomposition. (Ex 7.4) Principal Component Analysis (PCA)**	H.W #16	